Houxiang Ji

Contact Information

AI System Staff Engineer
SK hynix America
San Jose, CA

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Research Interests

Computer Systems and Architecture, Memory Systems, Machine Learning Systems, Processing-in/near-Memory, Emerging Memory, Compute Express Link (CXL), Hardware/Software Co-design

Education

University of Illinois Urbana-Champaign
Ph.D., Computer Science, 2025
Advisor: Prof. Nam Sung Kim
Thesis Committee: Prof. Nam Sung Kim (Chair), Prof. Josep Torrellas, Prof. Tianyin Xu, Prof. Saugata Ghose, Dr. Ren Wang (Intel)

University of Illinois Urbana-Champaign
M.Sc., Computer Science, 2021

Shanghai Jiao Tong University
B.Eng., Computer Science and Engineering, 2018
Zhiyuan Honor Degree in Computer Science
B.Sc., Finance and Economics, 2018

Professional Experience

Memory Systems Research, SK hynix America
AI System Staff Engineer — Aug 2025 – present

Capacity Engineering and Analysis, Meta
Visiting Researcher — Jun 2022 – Dec 2022

CPU Design, Luminous Computing
Research Intern — May 2021 – Aug 2021

Security and Privacy Research, Intel Labs
Research Intern — May 2020 – Aug 2020

Honors and Awards

- IEEE/ACM MICRO Hall of Fame, 2026
- SK hynix Memory Systems Research Gold Practice Award, 2026
- SK hynix Memory Systems Research Recognition Award, 2025
- Best Demo Award, SRC/DARPA PRISM Center Annual Review, 2023
- UIUC CS Ph.D. Fellowship, 2023
- Ray Ozzie Computer Science Fellowship, 2018
- Outstanding Graduate of Shanghai Jiao Tong University, 2018
- Student Travel Grants: USENIX ATC 2025, IEEE/ACM MICRO 2024, USENIX OSDI/ATC 2023, ACM/IEEE ISCA 2019

Publications

Conference Papers
First/co-first (top-tier): MICRO×4, ATC×2
All top-tier (18 of 23): MICRO×8, ISCA×2, ASPLOS×2, HPCA×2, OSDI×1, ATC×2, DAC×1
Artifact badges: Available Functional Reproduced

[C23] Jaeyoung Kang, Sujin Kim, Hyunji Kim, **Houxiang Ji**, Ji-Hoon Kim, Nam Sung Kim. *UPS: Unlocking On-Chip Analytics Accelerator for Memory-Efficient Filtered Proximity Graph Search.* **IEEE HPCA 2027**

[C22] Yan Sun, Taehyung Lee, Vinit Gupta, Austin Antony Cruz, Dongming Liu, **Houxiang Ji**, Ren Wang, Nam Sung Kim. *Don't Wait! Let CXL Deliver the Data for You.* **IEEE HPCA 2027**

[C21] **Houxiang Ji***, E. Ezgi Yücel*, Yaochen Li, Prajwal Ramakrishna, Derrick Quinn, Jungmin Choi, Hoshik Kim, Mohammad Alian, José F. Martínez. *C^3: Collaborative CPU-Commodity DRAM Computation for Reliable LLM Acceleration.* **IEEE/ACM MICRO 2026** *(* equal contribution)* **[Featured by SK hynix]**

[C20] Minho Kim, **Houxiang Ji**, Hwanjun Lee, Jaeyoung Kang, Sungju Kim, Hyunkyun Shin, Nam Sung Kim, Daehoon Kim. *AMG: AMX–GPU Cooperative Acceleration of Billion-Scale ANNS via GEMM Reformulation.* **IEEE/ACM MICRO 2026**

[C19] Yang Zhou*, **Houxiang Ji***, Bikrant Das Sharma, Jiyuan Zhang, Yu Li, Linjie Ma, Jaeyong Lee, Myoungjun Chun, Jihong Kim, Sudarsun Kannan, Nam Sung Kim. *CXL AnySSD: A Composable CXL SSD Using a CXL Type-2 Device and Any SSDs.* **IEEE/ACM MICRO 2026** *(* equal contribution)* **[Hardware prototype]**

[C18] Dusol Lee, Yan Sun, **Houxiang Ji**, Vinit Gupta, Austin Antony Cruz, Inhyuk Choi, Nam Sung Kim, Jihong Kim. *MAC: Metadata Acceleration for Sustainable Performance in Big-Data Systems with CXL DRAM.* **USENIX OSDI 2026**

[C17] Yifan He, Hyung Joon Byun, Jimin Lee, Do-Yoon Lim, **Houxiang Ji**, Jungmin Choi, Hoshik Kim, Jae-sun Seo. *PAGE: Processing-Using-DRAM Architecture-Circuits Co-Optimization for Efficient Acceleration of General Matrix-Vector Multiplication.* **ACM/IEEE DAC 2026**

[C16] **Houxiang Ji**, Yifan Yuan, Yang Zhou, Ipoom Jeong, Ren Wang, Saksham Agarwal, Nam Sung Kim. *Re-architecting End-host Networking with CXL: Coherence, Memory, and Offloading.* **IEEE/ACM MICRO 2025 [Hardware prototype]**

[C15] **Houxiang Ji**, Minho Kim, Seonmu Oh, Daehoon Kim, Nam Sung Kim. *Paraksm: Parallelized Memory Deduplication with Data Streaming Accelerator.* **USENIX ATC 2025**

[C14] **Houxiang Ji**, Srikar Vanavasam, Yang Zhou, Qirong Xia, Jinghan Huang, Yifan Yuan, Ren Wang, Pekon Gupta, Bhushan Chitlur, Ipoom Jeong, Nam Sung Kim. *Demystifying a CXL Type-2 Device: A Heterogeneous Cooperative Computing Perspective.* **IEEE/ACM MICRO 2024 [Featured by Intel]**

[C13] Jinghan Huang, Jiaqi Lou, Srikar Vanavasam, Xinhao Kong, **Houxiang Ji**, Ipoom Jeong, Danyang Zhuo, Eun Kyung Lee, Nam Sung Kim. *HAL: Hardware-assisted Load Balancing for Energy-efficient SNIC-Host Cooperative Computing.* **ACM/IEEE ISCA 2024**

[C12] Chihun Song, Michael Jaemin Kim, Tianchen Wang, **Houxiang Ji**, Jinghan Huang, Ipoom Jeong, Jaehyun Park, Hwayong Nam, Minbok Wi, Jung Ho Ahn, Nam Sung Kim. *TAROT: A CXL SmartNIC-Based Defense Against Multi-bit Errors by Row-Hammer Attacks.* **ACM ASPLOS 2024**

[C11] Yan Sun, Yifan Yuan, Zeduo Yu, Chihun Song, Reese Kuper, Jinghan Huang, **Houxiang Ji**, Siddharth Agarwal, Jiaqi Lou, Ipoom Jeong, Ren Wang, Jung Ho Ahn, Tianyin Xu, Nam Sung Kim. *Demystifying CXL Memory with Genuine CXL-Ready Systems and Devices.* **IEEE/ACM MICRO 2023 [300+ citations]**

[C10] **Houxiang Ji**, Yan Sun, Mark Mansi, Yifan Yuan, Jinghan Huang, Reese Kuper, Michael M. Swift, Nam Sung Kim. *STYX: Exploiting SmartNIC Capability to Reduce Datacenter Memory Tax.* **USENIX ATC 2023**

[C9] Lihui Liu, **Houxiang Ji**, Jiejun Xu, Hanghang Tong. *Comparative Reasoning for Knowledge Graph Fact Checking.* **IEEE BigData 2022**

[C8] Zhangxiaowen Gong, **Houxiang Ji**, Yao Yao, Christopher Fletcher, Christopher Hughes, Josep Torrellas. *Optimizing Graph Neural Networks on CPUs via Cooperative Software-Hardware Techniques.* **ACM/IEEE ISCA 2022**

[C7] Zirui Neil Zhao, **Houxiang Ji**, Adam Morrison, Darko Marinov, Josep Torrellas. *Pinned Loads: Taming Speculative Loads in Secure Processors.* **ACM ASPLOS 2022**

[C6] Zhangxiaowen Gong, **Houxiang Ji**, Christopher Fletcher, Christopher Hughes, Josep Torrellas. *SparseTrain: Leveraging Dynamic Sparsity in Software for Training DNNs on General-Purpose SIMD Processors.* **IEEE/ACM PACT 2020**

[C5] Zhangxiaowen Gong, **Houxiang Ji**, Christopher Fletcher, Christopher Hughes, Sara Baghsorkhi, Josep Torrellas. *SAVE: Sparsity-Aware Vector Engine for Accelerating DNN Training and Inference on CPUs.* **IEEE/ACM MICRO 2020**

[C4] Zirui Neil Zhao, **Houxiang Ji**, Mengjia Yan, Jiyong Yu, Christopher W. Fletcher, Adam Morrison, Darko Marinov, Josep Torrellas. *Speculation Invariance (InvarSpec): Faster Safe Execution Through Program Analysis.* **IEEE/ACM MICRO 2020**

[C3] **Houxiang Ji**, Li Jiang, Tianjian Li, Naifeng Jing, Jing Ke, Xiaoyao Liang. *HUBPA: High Utilization Bidirectional Pipeline Architecture for Neuromorphic Computing.* **ASP-DAC 2019**

[C2] **Houxiang Ji**, Linghao Song, Li Jiang, Hai (Helen) Li, Yiran Chen. *ReCom: An Efficient Resistive Accelerator for Compressed Deep Neural Networks.* **DATE 2018 [100+ citations]**

[C1] Luyu Li, **Houxiang Ji**, Chentao Wu, Jie Li, Minyi Guo. *Favorable Block First: A Comprehensive Cache Scheme to Accelerate Partial Stripe Recovery of Triple Disk Failure Tolerant Array.* **ICPP 2017**

Journal Papers

[J5] **Houxiang Ji***, Minho Kim*, Jaeyoung Kang, Hwanjun Lee, Daehoon Kim, Nam Sung Kim. *CABANA: Cluster-Aware Query Batching for Accelerating Billion-Scale ANNS with Intel AMX.* **IEEE Computer Architecture Letters (CAL)**, 2025 *(* equal contribution)*

[J4] Bikrant Das Sharma, **Houxiang Ji**, Ipoom Jeong, Nam Sung Kim. *Time Series Machine Learning Models for Precise SSD Access Latency Prediction.* **IEEE CAL**, 2025

[J3] **Houxiang Ji***, Qirong Xia*, Yang Zhou, Nam Sung Kim. *Hardware-accelerated*

Kernel-Space Memory Compression Using Intel QAT. **IEEE CAL**, 2025 *(* equal contribution)*

[J2] Chihun Song, Michael Jaemin Kim, Yan Sun, **Houxiang Ji**, Kyungsan Kim, TaeKyeong Ko, Jung Ho Ahn, Nam Sung Kim. *X-PPR: Post Package Repair for CXL Memory*. **IEEE CAL**, 2025

[J1] **Houxiang Ji**, Minho Kim, Seonmu Oh, Daehoon Kim, Nam Sung Kim. *Cooperative Memory Deduplication with Intel Data Streaming Accelerator*. **IEEE CAL**, 2025

Workshop Papers

[W1] Matteo Olivi, Aubhro Sengupta, Jimin Lee, **Houxiang Ji**, Jungmin Choi, Ada Gavrilovska. *Characterizing CXL Memory Management in a Production Cluster Management Host Agent*. **HCDS 2026**, co-located with ASPLOS 2026

Invited Talks

Demystifying a CXL Type-2 Device: A Heterogeneous Cooperative Computing Perspective
- Shanghai Jiao Tong University, Dec 2024
- OCP Global Summit, San Jose, CA, Oct 2024
- OCP Composable Memory Systems Workgroup, Mar 2024

Memory-Intensive Kernel Feature Acceleration with CXL Type-2 Devices
- OCP Global Summit, San Jose, CA, Oct 2023
- OCP Composable Memory Systems Workgroup, Oct 2023

Conference Presentations
- USENIX ATC 2025, IEEE/ACM MICRO 2024, USENIX ATC 2023

Professional Service

Technical Program Committee
HPCA 2027, MICRO 2026, DAC 2026/2027, DATE 2027, ISMM 2026, GLSVLSI 2025/2026, RTCSA 2026

Journal Reviewer
- Computer Architecture Letters (CAL)
- Transactions on Architecture and Code Optimization (TACO)
- Transactions on Emerging Topics in Computing (TETC)
- Transactions on Mobile Computing (TMC)
- Journal on Emerging Technologies in Computing Systems (JETC)

Artifact Evaluation Committee
ASPLOS 2025, OSDI 2024, ATC 2024, MICRO 2024